Millets Value Chain Platform
A Digital Marketplace for "Shree Anna"

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I. ABSTRACT

The nutritional value of the millets has provided them with a renewed significance in the world, along with their climate and environmental sustainability in agriculture. Nevertheless, the current millet value chain in India and other developing economies is highly disintegrated, and it is also defined by limited market accessibility, absence of price transparency, lack of supply chain coordination, and low digital integration as the recent studies on value chain dynamics, perception of consumers, and governance gaps within agri-food systems. Low bargaining power, insufficient certification mechanisms, and relying on intermediaries, in turn, hinder the potential of tribal and smallholder farmers to make an income and enter the market.

The proposed paper is a concept of a digital Millet Supply Chain Platform that will reduce all financial limits between farmers, startup processors, customers and administrators into a single online platform. The platform facilitates the process of direct-to-consumer and direct-to-startup transactions between farmers and consumers, online seller verification, raw material purchase, selling of processed goods, traceability, price transparency, an ACM (Accessories, Craft & More) waste-to-value system, and online order and payment tracking. Based on the international trends of digitalizing the agri-food system and traceability, the system can rely on the principles of a modular architecture, role-based dashboards, digital monitoring, and sustainability-focused value chain**.**

This strategy aims at integrating value chain optimization models, millet-based market knowledge, and consumer-centred product strategies to develop a technology-enabled supply chain. The result is a platform which improves the income of farmers, startup development, quality assurance, market transparency, traceability and sustainable millet-based value chains. The solution that has been proposed will eventually create an inclusive, effective, and digitally empowered agricultural market that is congruent with the future demands of the millet economy.

Keywords: Millet value chain, digital supply chain, agri-food traceability, online marketplace, sustainable agriculture, farmer-to-consumer platform.

A. Introduction

B. Background and Motivation

Today, the millets are again re-introduced to the world due to its rich nutritional value, low GI, drought-resistance, and the ability to grow in dry regions [12]. Millets also contribute significantly to sustainable agriculture and nutrition security due to their adaptability and health benefits [13].

Millets contribute to improving food security as stated in the literature that has been gathered in India, Sri Lanka among other markets in the world [1]. They also encourage sustainable farming and enhance the livelihood of the smallholder and tribal farmers [2]. Also, millets help in the economic and ecological stability within risky areas [11].

Millet products and their value-added product demand, whether in raw or value-added forms, are constantly rising thanks to the increasing consumer awareness [7]. The consumption of more healthy and nutritious food alternatives is also increasing [12].

The potential notwithstanding, there are a number of limitations in the millet market in India. The value chain is inefficient causing low profits to farmers [9]. Proper storage and processing facilities are also lacking and this has an impact on the quality of the products and the consistency of supply [8]. Additionally, the control of the middlemen and the lack of institutional backing have a detrimental effect on farmer income [9].

The studies on consumer perception indicate that there is poor availability and visibility of processed millet products in the market [7]. Moreover, the operations of the supply chain in developing economies are not mainly digitalized, and this is why there is poor traceability [3]. This leads to fragmented market relations and poor transparency in dealings [5].

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With these difficulties in mind, an online platform that would unite stakeholders and increase the visibility is highly demanded. This type of system can enhance traceability and offer direct access to the market [4]. The necessity of technology solutions is also supported by global attention to the digital governance and agri-food traceability [17]. These solutions have the potential to empower the millet farmers, startups, and consumers to develop a sustainable value chain [19].

C. Problem Definition and Objectives

There are a number of structural constraints to the current millet ecosystem. Price and market transparency are lacking and this decreases the bargaining power of farmers [1]. Fragmented supply chain networks do not allow coordination among farmers, processors and buyers to be effective [6]. Over reliance on intermediaries lowers the earnings of farmers and raises consumer prices [1]. The key issues within developing economies are also poor traceability and weak quality control systems [5]. Transparency and accountability is also restricted by the absence of digital infrastructure [16]. Platforms to sell by-products of agriculture are also scarce, which impacts on the possibilities of the circular economy [13]. The startup processors have difficulties in getting uniform quality raw materials at stable prices [11]. In order to solve these problems, the suggested system will seek to establish a single digital market place that will link all the stakeholders [15]. The platform will be transparent in pricing and systems of verified sellers [4]. It will facilitate raw and processed millet products with the aim of adding value [7]. The system will also have order management modules, payment tracking modules, certification verification modules and feedback modules [3]. It will also aid sustainability using ACM (Accessories, Craft and More) using agricultural waste [13]. On the whole, the idea is to establish a niche that will boost farmer incomes, startup growth, and consumer confidence [6].

D. Overview of the Proposed Project.

The Millet Supply Chain Platform proposed is a multi-stakeholder multi-functional digital platform. The farmers will get the opportunity to establish authenticated profiles, sell raw and by-product products, handle orders, collect payments, and interact with customers [6]. Businessmen are able to buy raw materials at the farms, advertise value added products of millet and handle orders and payments effectively [11]. Customers will be able to purchase products, see nutrition benefits, and communicate with the community by sharing recipes [7]. The administration will handle the registration, certification verification and track system activities and quality control [4]. The system combines various functions such as an online marketplace where millet products can be purchased, traceability, and certification validation. It also incorporates payment integration with the use of UPI systems, ACM modules covering the use of waste, and community interaction capabilities. It is also equipped with order tracking, delivery systems, and feedback mechanisms to provide transparency and efficiency [13]. In general, the aim of the platform is to fill the essential gaps in the current value chains through empowering farmers, rural startups and providing quality products to consumers [17].

E. Outline of the Paper

The rest of this paper is structured in the following way:

Section 3 is a broad literature review that dwells on millet value chains, digital agricultural supply chains, traceability frameworks, and research gaps found in the current systems.

Section 4 presents the methodology used in the proposed platform, which contains requirement analysis, system design strategy, development workflow, and testing process.

Section 5 outlines the system design and implementation, which offers the architectural framework, functional modules, data flow mechanisms, and implementation strategies.

Section 6 deals with the experimental findings and system performance with focus on usability, system efficiency and better performances in comparison to the traditional supply chain practices.

Section 7 will deal with the social, ethical and sustainability issues of the proposed digital platform.

Section 8 is the conclusion of the paper summarizing the significant contributions and research findings.

Section 9 contains references.

II. Literature Review

Various literature has shown that value chain that underpins millet farming and marketing is still poor and underdeveloped even though the crop is nutritionally balanced and resistant to climate changes. A study in the tribal area of Attappadi in Kerala indicates that middlemen in the area control the supply chain and reduce the bargaining power of farmers hence leading to low prices [1]. Additional international assessments indicate the potential and the difficulties of increasing millet markets and the necessity to enhance the structural support systems [2].

Digitization of agri-food systems (as opposed to the traditional ones) has been noted as a major step towards enhancing coordination, minimizing inefficiencies, and sustainable governance in emerging economies [3]. Comparative research in OECD countries shows that digital traceability systems lead to food safety, better monitoring of supply chains, and better governance mechanisms [4]. The developing importance of digital ledgers, cloud computing, and integrated databases in tracking the origin, prices, certifications, and movements of products along value chains is also further explained in literature on agri-food traceability [5].

Research on agricultural value chains in developing nations has highlighted the relevance of organized market connections, clear pricing approaches, and better

coordination towards increasing farmer well-being [6]. Karnataka regional studies indicate that the smallholder millet farmers have had disjointed markets, poor aggregation and access to organized buyers [7]. Farmer Producer Organizations (FPOs) have been brought forward as possible solutions to enhance the value chain but these are only accessible and effective to a limited and irregular extent [8].

Although there is rising consumer awareness, value-added millet products are still struggling with low visibility, a lack of consumer familiarity, and irregular supply chains [9]. Widening reviews on millet value chain development emphasize the necessity of having integrated strategies that have a stronger linkage between production, processing and marketing activities [10]. Studies of agribusiness development in the millet industry show that small processors and startups are faced with inconsistent access to raw material and inaccurate data on the market [11].

Nutritional research reports that millets have a major health advantage namely high nutrient density, and the ability to fit healthy diets, which explains the increasing popularity of the product with consumers [12]. The importance of millets in encouraging sustainable food systems is also highlighted in global reports because of their minimal water needs and climate change resilience [13]. Other government policies like the National Year of Millets also indicate that government has undertaken measures to encourage production and consumption of millets nationally [14].

Agricultural marketing has been identified to be an effective approach through digital platforms to enhance access to the market, price transparency, and less reliance on intermediaries [15]. There is also an exploration of emerging technologies like blockchain to improve on transparency, traceability, and trust in agricultural supply chains [16]. According to reports by international organizations, digital technologies are important in enhancing food supply chains and in general efficiency and governance [17].

Research on e-agriculture and digital marketplaces with a farmer focus suggests that this type of marketplace can close the divide between farmers and consumers by providing a direct market access and enhancing information exchange [18]. National-level strategies also highlight the need to have systematic strategies to advance millets and to build their value chains by policy and technological interventions [19]. Also, the development of smart farming and big data technologies proves that the data-driven approach can enhance the productivity of agriculture and its supply chain efficiency [20].

In sum, the literature demonstrates that despite their nutritional, and growing popularity, the millets value chain is inefficient at all levels of production, aggregation, processing, and consumption [2]. Conventional supply chains are not transparent, coordinated, and digitalized, which is a major impediment to their performance [15]. Simultaneously, international research studies show that efficiency, trust, and sustainability may be enhanced by implementing digital supply networks and traceability systems [17]. This provides a good reason to develop a digital Millet Supply Chain Platform that will combine the ideas of value chains with the current supply chain technologies and thus give a transparent market, enhance stakeholder connectivity, traceability, and empower the entire millet ecosystem [19].

III. Methodologies

The design of the proposed digital millet supply chain infrastructure has been developed in an organized system development methodology of practical feasibility, modular design, and iterative validation. The approach focused on the experiential study of practical agricultural issues and their conversion into practical system constraints and deployment of a working prototype to test usefulness and performance.

A. Requirement Analysis

The requirement analysis stage was meant to learn about the issues recorded in literature- broken market linkages, absence of transparency, frail traceability and the necessity to digitalize the system and convert them into system specifications. According to the needs of the stakeholders observed in the study of value chains [3], the requirements were separated into three categories: The aim of the requirement analysis phase was to be familiar with the problems that were identified in literature; i.e. broken market connection, lack of transparency, poor traceability, and need of digital integration and transform them into real functionality of the system. They were divided into three groups according to the needs based on the requirements of all the stakeholders present in the value chain analysis [10].

1. Functional Requirements

The system has various user roles that include farmers, customers, startup owners and administrators with respective functionalities in the platform. It offers product listing modules that facilitate processed products as well as by-products under the ACM category. The platform allows order management and payment gateways like UPI and cash on delivery, and delivery tracking. It also has the elements of verifying the profiles such as Aadhaar, which is certified by the government to verify authenticity. The system also contains a system of feedback and rating to increase trust and transparency among users. It promotes the sharing of recipes and communication platforms to ease the communication between the customers and other stakeholders. Moreover, administrative capabilities comprise approvals, activity tracking and analytics dashboard to monitor the overall system performance.

2. Non-Functional Requirements

The system is geared to provide a high level of efficiency even to rural users with low digital penetration so that it is easy to use and accessible. It includes secured and encrypted data storage and appropriate authorization systems to keep user data safe. The site will be designed to be scalable with responsive page loading to support growing users and data effectively. To be reliable, it guarantees a smooth flow of transactions without errors. The system also has a robust infrastructure of maintaining an accurate and consistent information.

3. Stakeholder Requirements

The platform will support the interest of various stakeholders, as farmers will be able to find transparent pricing, validated purchasers, and uncomplicated order tracking systems. A regular supply of raw materials and access to processed millet products are needed by the owners of the startups to carry out their business. Customers demand authentic products and trustworthy information about nutrition and the origin of the product. The administrators are in charge of the general management of the platform, verification process and monitoring activities, in order to assure the trust, transparency and smooth operation of the system.

B. System Design Approach

The system design is based on a modular, layered architecture, which is consistent with the ideas of the literature on digital supply chain [4].

1. *Layered Design*

   Presentation Layer- User Interface/User-Experience.

   Application Layer- Business logic of orders and payments and verification.

   Data Layer- Secure data flow in centralised database.

2. *Design Principles*

   The system is designed with the ability to support expansion, allowing an increasing number of farmers and startup owners to register and participate in the platform without affecting performance. It ensures interoperability so that certification information and traceability data can be efficiently integrated and shared across different modules. Sustainability is also a key consideration, with support for ACM-based waste-to-value models that promote effective utilization of agricultural by-products. Additionally, the platform follows a user-centered design approach that is simple and intuitive, making it accessible and usable for farmers with varying levels of literacy and digital familiarity.

3. *Workflow*

   Each of them (farmer, startup, customer, admin) is developed as a distinct functional component that is presented via a separate dashboard, yet linked to a central data structure that ensures a smooth interaction.

C. Development Process

The progression was done in a step-by-step workflow process that was useful in organizing the system in an efficient manner as well as giving a chance to make improvements at every stage.

1. Information Gathering and Requirement Study

This was initiated by gathering the extensive requirements based on the literature research, stakeholder consultations and on the response of our project reviewer. This assisted us in clarifying clearly what each user role, i.e. Farmer, Startup Owner, Customer and Admin, required of the system.

2. User Interface (UI) Design

After the requirements were clear, we proceeded on to the design of the user interface. All major modules such as product listing, dashboards, order pages, and admins had wireframes and screen layouts developed. The aim was to maintain the interface as simple, friendly to the mobile device, and easy to understand among the users with different degrees of digital literacy.

3. Backend Development and Integration

Backend was developed after the completion of the UI. The orders, payments, verification, and feedback business logic, the database structure, APIs and authentication modules were implemented. The system was then progressively linked to backend services by connecting each component of UI.

4. System Integration

Each of them was incorporated into one platform (farmer, startup, customer, admin, payments, and ACM). Data flow between modules was also tested to check out the consistency, ease of navigation and proper handling of transactions.

5. Testing and Refinement

There were several stages of testing which included: functional, usability, error handling and performance testing. Any identified problems were fixed, and the workflow could be streamlined to enhance the general usability particularly by farmers and those who were using digital equipment the first time.

D. Testing and Validation

During the development, testing was conducted to make sure that the platform was developed and functioning. Each module underwent basic functionality testing and the modules were then integrated together when the frontend and the backend have been linked. The usability was checked in order to ensure that the interface was easily navigated by farmers, buyers and administrators. Output validation was done through the examination of outputs, error correction, and feedback of project guide/reviewer.

E. Tools and Environment

The site has been built on a newer stack of web technology:

Frontend: HTML, CSS, JavaScript, React

Backend: Node.js (Express)

Database: MySQL

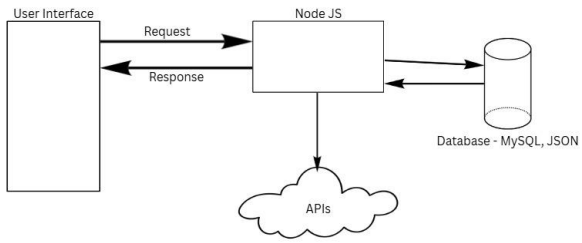
External Services: Google Translate API

Development Tools: VS Code, Git/GitHub

Operating Environment: Windows 10/11

IV. System Design and Implementation

A. System Architecture



The architecture of the system consists of three layers in order to provide modularity, scalability and simplicity to maintain:

Presentation Layer:
The user interface was developed with the help of React and offers farmers, consumer, and administrator pages.

Application Layer:
The functionality implemented on the backend with Node.js/Express and that services authentication, product management and orders, payments and the generation of certificates.

Data Layer:
MySQL database with user, product, transaction and certificate as well as information related to millet.

This layer independent architecture enables all the layers to be modified or scaled without any impact on the other parts of the system.

B. Functional Modules

1. Farmer Module

Farmer module, allows farmers to create an account and log-in to the platform, in order to view their own dashboard. Farmers will be able to advertise their millet products with the quantity and pricing information. The module will also enable them to track the performance of their products, orders, and any details that relate to payment, which will give them better control of their sales operations.

2. Consumer Module

The consumer module offers the registration and login services to the users to access the platform. It enables the customers to browse through a range of millet products and specifications of their nutritional and health values. The system allows consumers to make payments and place orders, thereby guaranteeing them a smooth purchasing experience.

3. Admin Module

The role of the admin module is to maintain the account of the farmers and consumers in the platform. It offers the feature of reviewing and approving of product listing, certificate validation and authenticity. Also, administrators keep track of the overall platform performance and control transactions and interactions to ensure consistency in the system.

4. Payment Module

The payment module enables safe e-commerce payments to consumers. It gives real-time information to farmers on the status of their payments and transparency in financial activities. The module is also intertwined with third-party and UPI-based payment gateways, which allows processing the payments in a convenient and reliable manner.

5. Certificate Module

The certificate module also increases the integrity and credibility of the value chain since it confirms and prints certifications on products and sellers. This will give confidence to the consumers and help in quality assurance in the platform.

6. Health Benefits Module

The health benefits module offers scientifically-backed information on the different millet varieties and their health benefits. It assists consumers to make sound dietary decisions by being aware of the health consequences of the products that they buy.

7. Government Schemes Module

The government schemes section gives current information regarding the policies and initiatives concerning millets. It assists the farmers to receive subsidies, training, incentives, hence contributing to their development and motivate them to join government-sponsored agricultural schemes.

C. Data Flow and Interaction

The system data flow will be drawn to make the database, which is handled by the back end and front end interacts with each other without any difficulty:

1. User Interaction:
   The requests are provided using the React interface by users (farmer/consumer/admin).

2. Backend Processing:
   The request is dispatched to the Node.js/Express server in which the inputs are checked and necessary operations are performed (authentication, product management, order creation, etc.).

3. Database Operations:
   The backend is updated or queried against the MySQL database when the need arises.

4. Response Handling:
   The result of the processing is directed back to the frontend and displayed to the user.

This stream will ensure that data is updated in real time, interaction is safe, and system behavior is not problematic.

D. Implementation Details

Frontend Implementation:
Components in react were crafted to accommodate each user class with pure UI designs. Forms, product cards, and dashboards were made to have reusable components.

Backend Implementation:
The login, product management, payment, and certificate handling API were designed in the form of RESTful using Node.js/Express.

Database Implementation:
Users, products, orders, payments as well as certificates were organized into MySQL tables in appropriate relationships.

Integration:
The frontend and backend interact with the backend and MySQL respectively by using API calls and SQL queries respectively.

Testing & Deployment:
Testing was done on each module and subsequently integrated.

V. Results and Discussions

A. Usability and User Experience

The created platform was piloted with sample users who constituted the farmers, consumers, and admins. The interface was discovered to be easy to navigate through, intuitive and simple. Farmers could easily post the products, and the consumers could view and order millets by using few procedures. The layout was clear, responsive and the various modules were organized that made the user experience enjoyable.

B. System Performance

The system worked effectively in normal usage conditions. MySQL database retrieval was quick and API reasonably constant when interacted with a few times. The loading of pages was low and the backend was able to process user authentication, upload products as well as process orders without delays. The modular design was useful in maintaining similarity in performance of various portions of the platform.

C. User Benefits

The platform offers several direct advantages to its users by enhancing accessibility and efficiency across the millet value chain. Farmers are provided with an online marketplace where they can showcase and sell their millet products without dependency on intermediaries. Consumers gain access to clear and reliable information regarding product quality, pricing, and health benefits, enabling them to make informed purchasing decisions. Administrators are able to efficiently monitor platform activities, verify listings, and ensure the overall reliability of the system. Overall, the platform improves visibility, builds trust, and enhances convenience for all stakeholders involved in the millet ecosystem.

D. Improvement over Existing System

The proposed platform introduces significant improvements over the traditional millet marketing system, which is largely offline, unorganized, and dependent on middlemen. It enables direct interaction between farmers and consumers, thereby reducing exploitation and increasing fair pricing. The integration of online traceability enhances transparency in terms of product quality and pricing. The platform also provides easy access to government schemes and certifications, which are otherwise difficult to obtain through traditional channels. Additionally, centralized information on health benefits supports better consumer awareness and decision-making. Faster transactions and improved communication are facilitated through the digital system, making the overall process more efficient. These advancements demonstrate that the platform not only digitizes the value chain but also enhances its reliability, transparency, and farmer-centric approach.

VI. Social, Ethical and Sustainability Considerations

A. Social Impact

The social benefits of having the millet value chain platform are likely to be enormous. The system will address the issue of dependency on middlemen by allowing direct farmer entry to digital markets, which will contribute to the balanced income distribution. It empowers small and marginal farmers particularly rural and tribal areas through better market accessibility, visibility and bargaining power. Consumers also enjoy better transparency on origin of products, quality as well as price which results in generation of trust and informed decision making.

B. Ethical Considerations

The responsible information treatment of user data and open system practices would be considered an ethical way of implementing the platform. The data about the user (contact details, records of transactions and product data) should be stored and used with the legitimate purposes. The platform is not designed to be discriminatory and biased and offers equal access to all the groups of farmers without regards to their regions, gender, or the magnitude of production. Good communication on certifications, quality of the products, and pricing is ensured to avoid miscommunication or unethical marketing.

C. Sustainability Implications

Some of the known ecological benefits of millets include, low water needs, resistance to harsh climatic conditions, and the

ability to fit well in sustainable farm systems. The platform indirectly leads to environmental sustainability by encouraging people to consume millet and making markets linked effectively. The electronic system also saves the necessity of physical intermediaries, inefficiency in transportation, and paperwork. Promotion of value-added millet products and promotion of sustainable practices of farmers also resonates with other Sustainable Development Goals (SDGs), especially SDG 2 (Zero Hunger), SDG 12 (Responsible Consumption and Production), and SDG 8 (Decent Work and Economic Growth).

VII. Conclusion

The suggested millet value chain platform offers a systematic and technology-oriented solution to the predicaments that have endured in the millet market. The system allows farmers, consumers, and administrators to be connected to one digital system which increases transparency, enhances accessibility and lowers dependency on intermediaries. The modular system, as well as convenient interface, make sure the basic features of the system, including product listing, payments, certification display and access to government schemes, are effectively delivered.

The findings show that the platform is not only efficient in facilitating transactions but also encourages informed decision-making as products are well traced and their information made clear. This would mean increased market visibility and fair prices to the farmers and trust and convenience to consumers. In a wider sense, the system helps the social equity, ethical use of data and sustainability by promoting the production of millet that is resistant to climate changes and low-impact consumption.

All in all, the project proves that the digital intervention may become a huge boost to the millet value chain and a viable template that can be used to build upon in the future, such as the integration with mobile, sophisticated analytics, and additional services to support farmers.

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